AI 101 for Mechanical Engineers · Module 4

Learning with and without labels

Lesson 4.1

What you will learn

> Distinguish supervised and unsupervised learning from the feedback available, identify regression and classification tasks, and explain why an anomaly alert requires investigation.

Lesson overview

Module 4 · Lesson 1

Learning with and without labels

AI 101 for Mechanical Engineers

1 Supervised learning

Training examples contain inputs and known targets.

Inputs + known targets → Train a model

Regression	Classification
Predicts a continuous number, such as tool wear (mm)	Predicts a category or class probability, such as intact or cracked

Example: predict power from operating measurements, using measured power as the known target.

The target defines the task. The inputs can be numbers or images.

2 Unsupervised learning

Finds structure without supplied target labels for the task.

Operating profiles → Group by similarity

Groups may reflect load regimes, machine types or sensor drift.

Like sorting unlabelled parts by similarity, rather than studying with an answer key.

Interpret groups using measurement context and engineering evidence.

A cluster is not automatically a fault category.

3 Anomaly detection

Flags behaviour that differs from what is expected.

Unusual vibration or current

↓

Investigate the cause

↓

Check independent evidence

Possible causes: changed load, control settings, sensor problems or a fault.

Evidence: physical inspection, functional tests or calibrated reference measurements.

An alert alone does not confirm damage.

4 Semi-supervised learning

Uses a smaller labelled set together with additional unlabelled examples.

More unlabelled data do not guarantee improvement.

Key takeaway — **Identify the feedback first. Use engineering evidence to interpret groups and alerts.**

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Supervised and unsupervised learning use different feedback

Supervised

- Inputs paired with known targets [16]
- e.g. power from operating measurements
- Like practice questions with known answers

Unsupervised

- Finds structure without task target labels [16]
- Clustering groups by a similarity criterion
- Like sorting unlabelled parts by similarity

Unlabelled groups do not become confirmed fault classes without further evidence.

The target decides regression or classification

Regression

- Predicts a continuous numerical value
- e.g. tool wear (mm), pressure drop (Pa)

Classification

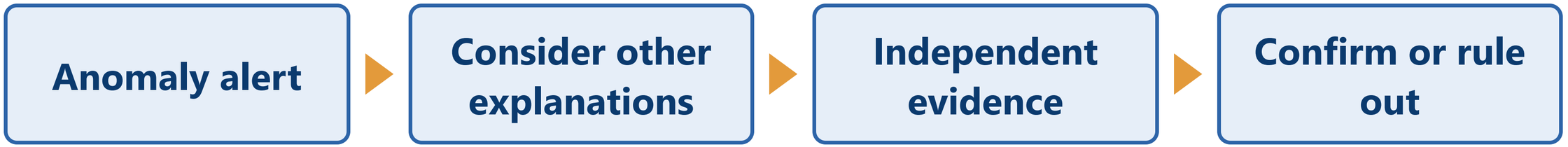
- Predicts a category or class probability
- e.g. intact versus cracked

> Numerical inputs or image inputs can support either task.

A cluster is not a health category

- Grouping operating profiles can reveal load regimes or machine types
- Similar observations may share a regime or sensor drift
- They are not automatically a confirmed fault
- Interpret groups with measurement context and engineering evidence

An anomaly alert starts an investigation



- Anomaly detection flags departures from expected behaviour [27]
- Possible causes: changed load, control settings, sensor problems, actual fault
- Evidence: physical inspection, functional testing, independent reference measurements

Semi-supervised learning mixes labelled and unlabelled data

- A smaller labelled set plus additional unlabelled examples [28]
- Can help when labels are costly
- Unlabelled data do not automatically improve a model
- Their distribution and the method’s assumptions must fit the task

> Keep the labelled reference and evaluation procedure explicit.

CHECK YOUR UNDERSTANDING

What distinguishes regression from classification?

Answer before moving on.

CHECK YOUR UNDERSTANDING

What distinguishes regression from classification?

> Regression has a continuous numerical target; classification has a category or class-probability output. The input format does not determine the distinction.

CHECK YOUR UNDERSTANDING

What additional evidence can turn an anomaly alert into a confirmed fault label?

Answer before moving on.

CHECK YOUR UNDERSTANDING

What additional evidence can turn an anomaly alert into a confirmed fault label?

> Independent evidence, such as physical inspection, functional testing or calibrated reference measurements, to confirm the fault and rule out other explanations.

Key takeaways

- Supervised learning uses input-target pairs
- Unsupervised learning finds structure without task-specific target labels
- Clusters require engineering interpretation
- An anomaly alert supports investigation; confirmation needs more evidence

References

[16] G. James, D. Witten, T. Hastie, and R. Tibshirani, An Introduction to Statistical Learning: With Applications in R, 2nd ed. New York, NY, USA: Springer, 2021. doi: 10.1007/978-1-0716-1418-1

[27] V. Chandola, A. Banerjee, and V. Kumar, “Anomaly detection: A survey,” ACM Comput. Surv., vol. 41, no. 3, pp. 1–58, 2009. doi: 10.1145/1541880.1541882

[28] O. Chapelle, B. Schölkopf, and A. Zien, Eds., Semi-Supervised Learning. Cambridge, MA, USA: MIT Press, 2006. doi: 10.7551/mitpress/9780262033589.001.0001

Numbers match the course reference list. The written lesson is the authoritative text.